Chapter 45

Analog I2C Controller

45.1 Introduction

The Analog I2C Controller contains two I2C masters dedicated to configuring and communicating with certain analog circuit modules. Those modules can adjust their operating states by configuring their internal registers to better suit the current application scenario of the chip. For example, the Analog I2C Controller can configure an analog sensor to add a uniform offset to the ADC sampling values, thereby extending the sampling voltage range. Each configurable analog module has an I2C slave internally, which is assigned with an independent address. The Analog I2C Controller supports the configuration of the following analog modules:

- CPU_PLL: This mainly includes the configuration register for CPLL_CLK, with a slave address of 0x67. For more information on CPLL_CLK, refer to Chapter 10 *Reset and Clock*.
- SYS_PLL: This mainly includes the configuration register for SPLL_CLK, with a slave address of 0x66. For more information on SPLL_CLK, refer to Chapter 10 *Reset and Clock*.
- MSPI: This mainly includes the configuration register for MPLL_CLK , with a slave address of 0x63. For more information on MPLL_CLK, refer to Chapter 10 *Reset and Clock*.
- PLLA: This mainly includes the configuration register for APLL_CLK , with a slave address of 0x6f. For more information on APLL_CLK, refer to Chapter 10 *Reset and Clock*.
- ANA_SENSOR: This mainly includes the configuration register for analog sensors (such as *Temperature Sensor (TSENS)* and *ADC Controller (ADC)*), with a slave address of 0x69.
- BIAS: This mainly includes the voltage detection configuration register, with a slave address of 0x6a. For more information, refer to Chapter 23 *Brown-out Detector*.

45.2 Feature List

The Analog I2C Controller has the following features:

- Master mode only
- 7-bit addressing
- Adjustable transmission rate
- Communication in the sleep modes supported by the Low-Power CPU
- Dual master operation mode

45.3 Architectural Overview

The figure below shows the architecture of the Analog I2C Controller.

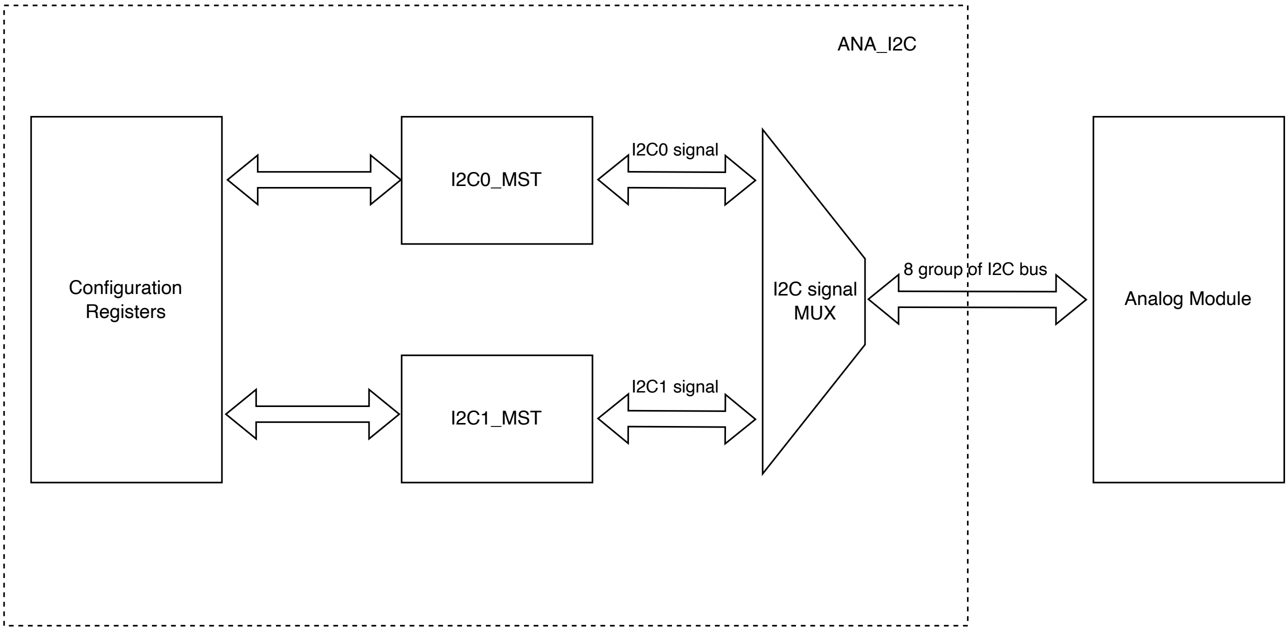

Figure 45.3-1. Analog I2C Controller Architecture

- Configuration Registers: A register module interfacing with the software, providing relevant configurations for the operation of the I2C master. This module controls the transmission address, data, etc., and monitors this transmission process by reading the status registers of I2C0/1_MST.
- I2C0/1_MST: Two I2C masters that can independently configure analog modules in parallel. The I2C signals they generate will be selected by a subsequent MUX and output to the analog modules.
- I2C signal MUX: A module that determines which analog master, I2C0_MST or I2C1_MST, configures a particular analog module.

45.4 Functional Description

Transmission Rate Adjustment

The transmission rate and signal phase of the I2C signal can be configured by ANA_I2C_MST_I2Cx_SDA_SIDE_GUARD and ANA_I2C_MST_I2Cx_SCL_PULSE_DUR. The relationship between the register values and the waveform is shown in the following figure.

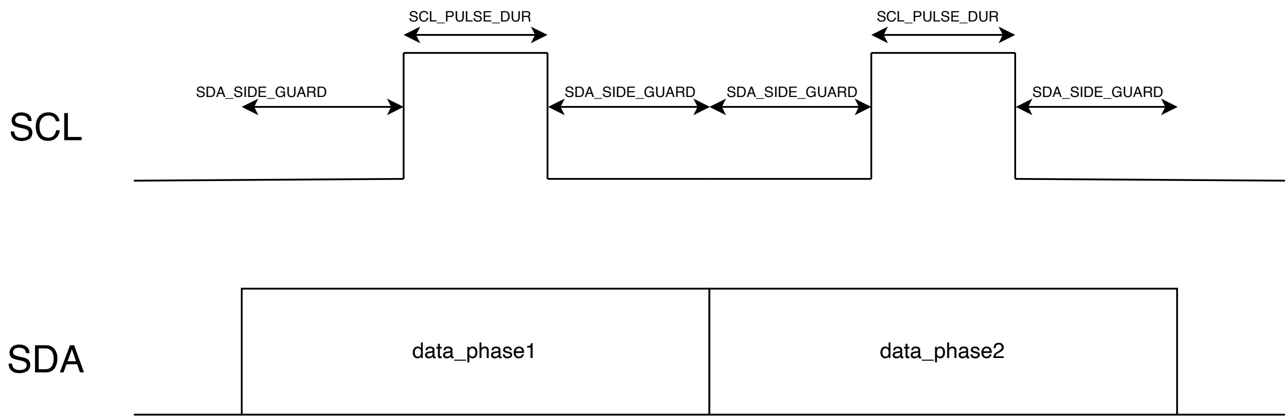

Figure 45.4-1. Signal Phase

Operation in Sleep Modes

In chip sleep modes, the operation of the Analog I2C Controller can be driven by the Low-Power CPU (see Chapter 3 *Low-Power CPU*).

Dual Master Operation Mode

The module integrates two independent master modules, each with its own set of configurations. The two masters can operate independently, configuring different analog modules, thereby improving operational efficiency.

45.5 Programming Procedures

The following is the process for configuring analog modules using the Analog I2C Controller.

- Write 1 to LPPERI_CK_EN_LP_I2CMST to enable the clock of the Controller.
- Configure ANA_I2C_MST_ANA_CONF2 to select the module for communication.
- Configure ANA_I2C_MST_I2C*x*_SDA_SIDE_GUARD and ANA_I2C_MST_I2C*x*_SCL_PULSE_DUR to determine the transmission rate and signal phase of the I2C signal.
- Configure ANA_I2C_MST_I2C*x*_CTRL to select relevant information for the current communication.
- Read ANA_I2C_MST_I2C*x*_BUSY. A low state indicates the end of the current communication for I2C*x*.

45.6 Register Summary

The addresses in this section are relative to Analog I2C Controller base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Configuration Registers			
ANA_I2C_MST_I2C0_CTRL_REG	I2C0 transmission configuration register	0x0000	varies
ANA_I2C_MST_I2C1_CTRL_REG	I2C1 transmission configuration register	0x0004	varies
ANA_I2C_MST_ANA_CONF2_REG	I2C master selection register	0x0020	varies
ANA_I2C_MST_I2C0_CTRL1_REG	I2C0 transmission rate and signal phase configuration register	0x0024	R/W
ANA_I2C_MST_I2C1_CTRL1_REG	I2C1 transmission rate and signal phase configuration register	0x0028	R/W
ANA_I2C_MST_DATE_REG	Version control register	0x0038	R/W

45.7 Registers

The addresses in this section are relative to Analog I2C Controller base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program registers that contain reserved fields, please refer to Section Programming Reserved Register Field.

Register 45.1. ANA_I2C_MST_I2C0_CTRL_REG (0x0000)

(reserved)	ANA_I2C_MST_I2C0_BUSY	ANA_I2C_MST_I2C0_CTRL	
31 26	25	24 0	
0 0 0 0 0 0	0	0x00000	Reset

ANA_I2C_MST_I2C0_CTRL Configures the transmission information for I2C0.
Bit[0:7]: Configures the slave address
Bit[8:15]: Configures the slave register address
Bit[16:23]: Configures the transmitted data
Bit[24]: Configures the read or write operation
0: Write
1: Read
Once this register is configured, the I2C0 master will generate I2C read or write signals.
(R/W)

ANA_I2C_MST_I2C0_BUSY Represents whether I2C0 is currently transferring data.
0: I2C0 is not transferring data
1: I2C0 is transferring data
(RO)

Register 45.2. ANA_I2C_MST_I2C1_CTRL_REG (0x0004)

(reserved)	ANA_I2C_MST_I2C1_BUSY	ANA_I2C_MST_I2C1_CTRL	
31 26	25	24 0	
0 0 0 0 0 0	0	0x00000	Reset

ANA_I2C_MST_I2C1_CTRL Configures the transmission information for I2C1. Its subfields have the same meaning with ANA_I2C_MST_I2C0_CTRL. (R/W)

ANA_I2C_MST_I2C1_BUSY Represents whether I2C1 is currently transferring data
0: I2C1 is not transferring data
1: I2C1 is transferring data
(RO)

Register 45.3. ANA_I2C_MST_ANA_CONF2_REG (0x0020)

reserved	ANA_I2C_MST_ANA_CONF2	
31 24	23 0	
0x0	0x0000	Reset

ANA_I2C_MST_ANA_CONF2 Configures which I2C master the following analog modules uses. If the corresponding bit is set to 1, I2C0 master is used for communication; if set to 0, I2C1 master is used. Since the slave address has already been defined, erroneous communication will not be caused.
Bit5: SYS_PLL
Bit6: SDIO_PLL
Bit7: ANA_SENSOR
Bit8: PLLA
Bit9: MSPI
Bit10: DIG_REG
Bit11: CPU_PLL
Bit12: BIAS
(R/W)

Register 45.4. ANA_I2C_MST_I2C0_CTRL1_REG (0x0024)

	(reserved)	ANA_I2C_MST_I2C0_SDA_SIDE_GUARD	ANA_I2C_MST_I2C0_SCL_PULSE_DUR	
Bits	31 … 11	10 … 6	5 … 0	
	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0x1	0x2	Reset

ANA_I2C_MST_I2C0_SCL_PULSE_DUR Configures the duration of the high-level period of the SCL driven by I2C0. The duration is measured by the counter operating on the LP_FAST_CLK clock. (R/W)

ANA_I2C_MST_I2C0_SDA_SIDE_GUARD Configures the duration of the low-level period of the SCL driven by I2C0. The duration is measured by the counter operating on the LP_FAST_CLK clock. (R/W)

Register 45.5. ANA_I2C_MST_I2C1_CTRL1_REG (0x0028)

	(reserved)	ANA_I2C_MST_I2C1_SDA_SIDE_GUARD	ANA_I2C_MST_I2C1_SCL_PULSE_DUR	
Bits	31 … 11	10 … 6	5 … 0	
	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0x1	0x2	Reset

ANA_I2C_MST_I2C1_SCL_PULSE_DUR Configures the duration of the high-level period of the SCL driven by I2C1. The duration is measured by the counter operating on the LP_FAST_CLK clock. (R/W)

ANA_I2C_MST_I2C1_SDA_SIDE_GUARD Configures the duration of the low-level period of the SCL driven by I2C1. The duration is measured by the counter operating on the LP_FAST_CLK clock. (R/W)

Register 45.6. ANA_I2C_MST_DATE_REG (0x0038)

(reserved)		Reserved	ANA_I2C_MST_DATE		
31	29	28	27	0	
0 0	0	0	0x2201300		Reset

ANA_I2C_MST_DATE Version control register. (R/W)